

Routine and Public Health Information Systems: Data for Public Health Practice

Keri Althoff, PhD
Johns Hopkins University



Except where otherwise noted, this work is available under the Creative Commons Attribution-NonCommercial-ShareAlike License 4.0

Objectives

- To introduce the principles and technical content of Routine (RHIS) and Public Health (PHIS) Information Systems
 - ▶ To define and describe routine and public health information systems
- Evaluate the vital registration system in countries in North, Central, and South America

Supporting Readings

- Mahapatra, *et al.* Civil registration systems and vital statistics: Successes and missed opportunities. Lancet 2007;370:1653-1663



Routine Health Information Systems: Civil and Vital Registration Systems

“Look for people who will aim for the remarkable, who will not settle for the routine.”

~David Ogilvy

Routine Health Information Systems (RHIS)

- Routine Health information systems produce timely, quality information for decision-making
- RHIS data are used to monitor progress towards goals, including:
 - ▶ Millennium Development Goals and Beyond
 - ▶ Health People (US)
- Traditional RHIS
 - ▶ **Civil** registration and **vital statistics** systems
 - ▶ **Medical** record systems
 - ▶ **Administrative** record systems (e.g. Census, Medicare)

Civil and Vital Registration Systems

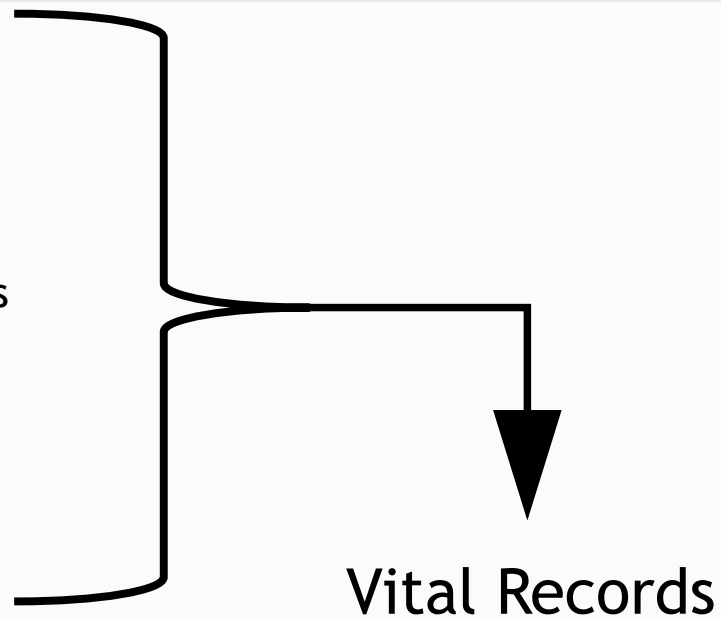
- China:
 - ▶ Zhou dynasty 1100-771 BC: Household registration system
- France
 - ▶ 1406: Archbishop of Nantes (secular after 1792)
- England and Wales
 - ▶ 1532: Bills of mortality
 - ▶ 1538: Civil registration system
 - ▶ 1850's: Vital statistics - William Farr

Civil Registration and Vital Statistics System

- USA
 - ▶ 1641: First attempt in Massachusetts
 - ▶ 1909: Compulsory for “registration states”
 - ▶ 1930: Completed in all states
 - ▶ 1950: US Vital Statistics System

Vital Registration Systems

- No registration system
- Verbal autopsy
- Sentinel registration system
- Demographic surveillance systems
- Sample registration systems
- Partial registration systems
- Full registration system



Vital Records

- “Certificates of birth, death, marriage and divorce required for legal and demographic purposes. Literally, records ‘to do with living.’”
- In the US (*full vital registration system*):
 - ▶ The state keeps these records
 - ▶ The federal government is also given some of the data collected from these records (http://www.cdc.gov/nchs/nvss/about_nvss.htm)
- World-wide
 - ▶ Records can be kept at the local, state, or federal level
- What data are collected from the vital records?

Fundamental Data Collected for Birth Certificate

- Baby
 - ▶ Date of birth
 - ▶ Place of birth (establish nationality)
 - ▶ Sex
- Mother
 - ▶ Date of birth
 - ▶ Race/ethnicity
 - ▶ Residence
 - ▶ Marital status at time of delivery
- Father
 - ▶ Date of birth
 - ▶ Race/ethnicity
 - ▶ Residence

"...it's a small paper but it actually establishes who you are and gives access to the rights and the privileges, and the obligations, of citizenship"

- Archbishop Desmond Tutu,
February 2005

Fundamental Data Collected for the Death Certificate

immediate

underlying

contributing

INTERNATIONAL FORM OF MEDICAL CERTIFICATE OF CAUSE OF DEATH		
Cause of death		Approximate interval between onset and death
I Disease or condition directly leading to death*	(a)	
	due to (or as a consequence of)	
Antecedent causes Morbid conditions, if any, giving rise to the above cause, stating the underlying condition last	(b)	
	due to (or as a consequence of)	
	(c)	
	due to (or as a consequence of)	
	(d)	
II Other significant conditions contributing to the death, but not related to the disease or condition causing it		
		

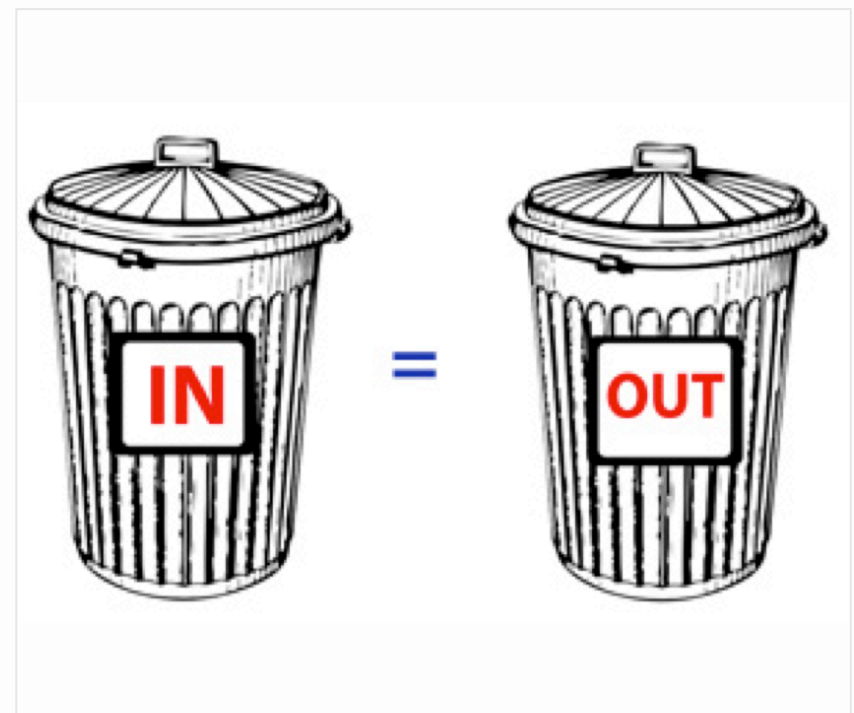
*This does not mean the mode of dying, e.g. heart failure, respiratory failure. It means the disease, injury, or complication that caused death.

These data are collected using International Classification of Disease version 10 (ICD-10) codes

International Statistical Classification of Diseases and Related Health Problems. Tenth Revision. 2nd ed, vol 2. http://www.who.int/classifications/icd/ICD-10_2nd_ed_volume2.pdf

Evaluating Civil Registration Systems

- Data from these systems are not perfect
 - ▶ (Mahapatra, *et al.* Civil registration systems and vital statistics: Successes and missed opportunities. Lancet 2007;370:1653-1663)



Accuracy of Data from Civil Registration Systems 1

- Coverage
 - ▶ % of population covered by medical certification of cause of death
- Completeness
 - ▶ % deaths with medically-certified cause of death
- Missing data
 - ▶ % of cause-of-death reports for which age, sex, or race data are missing

Accuracy of Data from Civil Registration Systems 2

- Use of ill-defined categories
 - ▶ % of deaths classified under various miscellaneous and ill-defined categories
- Improbable classifications
 - ▶ Number of deaths assigned to improbable age or sex categories, per 100,000 coded deaths
- Consistency between cause of death and general mortality
 - ▶ % of cause-of-death data points deviating more than 2 (or 3) standard deviations from general mortality based predictions



SECTION B: Routine Health Information Systems: Medical Record Systems

“Look for people who will aim for the remarkable, who will not settle for the routine.”

~David Ogilvy

Medical Record Systems

- These data are collected for clinical management of patients
 - ▶ This is not the same goal as data collection examining public health problems
 - ▶ Think about how data are collected in a medical record
 - Text from a physician
 - Lab values (internal and external laboratories)
 - Ordering tests and prescribing medicines
 - Rule-out diagnoses

Challenges When Using Medical Record Data

- Outcomes often need validation with external sources
 - ▶ Death registries
 - ▶ Cancer registries
- Changes in diagnosis or procedural codes must be considered
- Often the medical record system is not consistently used in a single hospital setting
- Electronic medical records are often focused on billing
- If the medical record system is not electronic
 - ▶ Handwriting issues with entry of data into an electronic format
 - ▶ Missing records
 - ▶ Abstraction and coding to put data into a usable form

Limitations to Medical Record Data

- Limitations to these data

- ▶ Measurement

- Behavioral data is usually not available
 - Timing of when information is collected (cigarette smoking status is usually not noted by physician at EVERY visit) ?
 - Data may not be complete if care is received elsewhere

- ▶ Study population

- Captures only those who are receiving health care
 - What is the denominator?



SECTION C: Routine Health Information Systems: Administrative Record Systems

“Look for people who will aim for the remarkable, who will not settle for the routine.”

~David Ogilvy

Administrative Record Systems

- This data can be very helpful in providing insight into the burden of disease in the population
- Commonly used sources of administrative records in public health
 - ▶ Census
 - ▶ Government-sponsored health programs (e.g., Medicaid & Medicare)
 - ▶ Insurance/Pharmaceutical databases (private)
- Commonly used administrative records in health statistics
 - ▶ Census data to determine the denominator

Challenges When Using Administrative Records

- This data is collected for administrative purposes, not for public health or statistic-generating purposes
- There can be barriers associated with using this data
 - ▶ Access - there is often no obligation to make data available
 - ▶ Financial costs
 - ▶ Time costs (applying for the data, gaining access, training as to how to use the data)

Limitations to Administrative Record Data

- Limitations to these data:
 - ▶ Measurement
 - Behavioral data are usually not available
 - Timing of when information is collected may not be consistent
 - Use of administrative/billing codes
 - ▶ Study population
 - Captures only those who are receiving benefits
 - What is the denominator?

RHINO

- RHINO “is a collaboration of developing country governments, donor agencies, technical groups, and private voluntary organizations whose purpose is to strengthen the role of evidence-based decision-making by engaging organizations and professionals in promotion of effective collection and use of routine health information, especially at the district level and below.”
- <http://rhinonet.org/index.php>



PRISM

- PRISM: Performance of Routine Information System Management
- Tools for Assessing, Monitoring, and Evaluating RHIS Performance
- <http://www.cpc.unc.edu/measure/tools/monitoring-evaluation-systems/prism>

Performance of Routine Information Systems Management (PRISM) Tools



SECTION D: Assessing the Quality of Mortality Statistics in the Americas

An Exercise

Quality of Mortality Statistics Index Score I

- PAHO and WHO have a quality index to assess mortality data from countries in the Americas
 - ▶ How well the registration system captures deaths
 - ▶ How well the underlying cause of death is characterized

- In the spreadsheet:
 - ▶ *Under-registered deaths (%)*: % of deaths which remain unregistered
 - ▶ *Ill-defined COD (%)*: % of deaths reported that listed a sign, symptom, or other ill-defined cause as the underlying cause of death

Quality of Mortality Statistics Index Score II

$$\text{Score} = [0.7 * \% \text{ of under-registered deaths}] \\ + [0.3 * \% \text{ ill-defined COD}]$$

- Data quality categories:
 - ▶ ≤10%: Good
 - ▶ 11-20%: Medium
 - ▶ 21-40%: Poor
 - ▶ ≥41% Very poor
 - ▶ No data = missing values for either under-registered or ill-defined causes

Quality of Mortality Statistics Index Score III

Technical notes:

- For some countries, the proportion of ill-defined or under registered deaths be missing. Countries with missing data for either should be placed in the “No data” data quality category.
- We are using a single year here to simplify the calculations. It is usually better, however, to use multiple years of data from each country to generate estimates of the quality of mortality statistics index score. The proportion of unregistered or ill-defined causes of death can vary from year to year for each country, so using multiple years of data produces estimates that more accurately reflect the performance of a country’s vital registration system.

Question 1 - Question

- Why do you think the data quality index places a greater weight on under-registered deaths as compared with ill-defined underlying cause of death?

WORK IT OUT >>

Question 1 - Answer

- Why do you think the data quality index places a greater weight on under-registered deaths as compared with ill-defined underlying cause of death?
- Underestimating mortality has a greater impact on mortality rates than ill-defined causes of death

Mortality rate = # of deaths / # in the population

Question 2 - Question

- In the spreadsheet, calculate the quality score for all of the countries. Which countries received a score of “good quality”? Which countries have the 3 lowest quality of data scores? (Hint: Use the sort data function)

WORK IT OUT >>

Question 2 - Answer

- In the spreadsheet, calculate the quality score for all of the countries. Which countries received a score of “good quality?” Which countries have the 3 lowest quality of data scores? (Hint: Use the sort data function)

Good Quality	
Antigua & Barbuda	Mexico
Argentina	Puerto Rico
Bahamas	Trinidad & Tobago
Bermuda	United States
Bolivia	Uruguay
Canada	Venezuela
Chile	Virgin Islands (UK)
Dominica	Virgin Islands (US)
Granada	

Lowest Quality Data	
Nicaragua	25.2%
Turks & Caicos	26.5%
Paraguay	27.0%

Question 3 - Question

- Consider the quality scores of Argentina and Bahamas. Both countries have “good quality” mortality data, however, they have different proportions of ill-defined COD and under-registered deaths. How is your interpretation of the mortality data from these two countries different?

WORK IT OUT >>

Question 3 - Answer

- Consider the quality scores of Argentina and the Bahamas. Both countries have “good quality” mortality data, however, they have different proportions of ill-defined COD and under-registered deaths. How is your interpretation of the mortality data from these two countries different?

	Quality Score	Ill-defined COD (%)	Under-registered deaths (%)
Argentina	5.9%	8.6%	4.7%
Bahamas	7.7%	1.7%	10.2%

Question 4 - Question

- Think of at least one reason why cause of death could be under-reported or misreported on a death certificate for HIV, homicide, and cancer deaths.

WORK IT OUT >>

Question 4 - Answer

- Think of at least one reason why cause of death could be under-reported or misreported on a death certificate for HIV, homicide, and cancer deaths.
- HIV:
 - ▶ Under-reporting: lack of HIV testing may result in people dying from HIV but a lack of knowledge of their infection
 - ▶ Misreporting: HIV stigma may results in intentional under-reporting
- Homicide:
 - ▶ Misreporting: People brought to the hospital with the family suggesting suicide, but perhaps it was homicide
- Cancer:
 - ▶ Under-reporting: lack of cancer diagnosis tests may result in people dying from cancer but a lack of knowledge of their diagnosis

Questions 5-7

Imagine that there is a new initiative to reduce the infant mortality rate in the countries where it is estimated to be the highest:

- Belize: 17.9 per 1,000 live births
- El Salvador: 21.5 per 1,000 live births
- Guatemala: 30.0 per 1,000 live births

Your role will be to evaluate the success of the new initiative to reduce the infant mortality rate.

Questions 5 - Question

- Will data supplied by the vital registration systems from Belize, El Salvador, and Guatemala accurately count the number of infant deaths (the numerator of the infant mortality rate)?

WORK IT OUT >>

Questions 5 - Answer

- Will data supplied by the vital registration systems from El Salvador, Belize, and Guatemala accurately count the number of infant deaths (the numerator of the infant mortality rate)?

	Under-registered deaths (%)	Quality of Mortality Statistics Index Score	Data Quality Category
Belize	31.9%	23.0%	poor quality
El Salvador	24.7%	22%	poor quality
Guatemala	12.7%	11%	medium quality

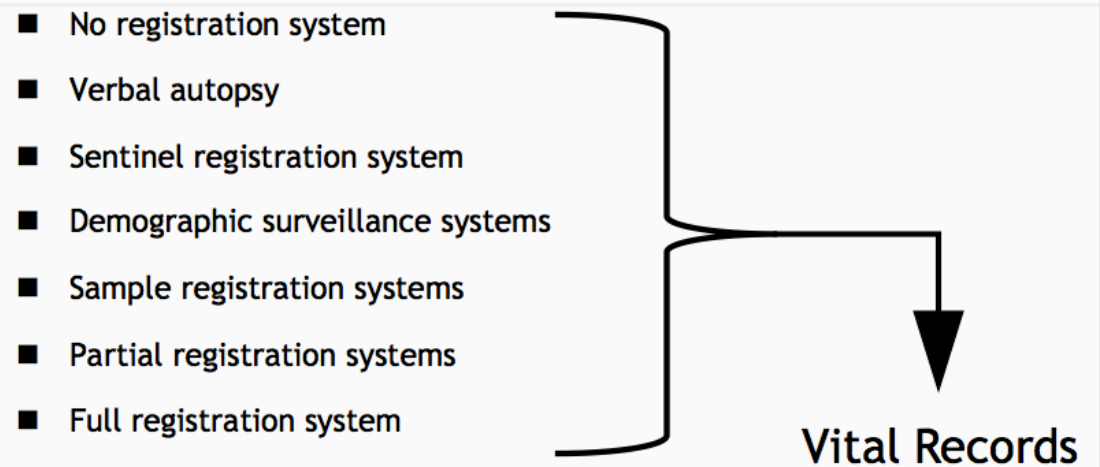
Question 6 - Question

- What other sources of data might you use to evaluate whether or not the infant mortality initiative reduced the infant mortality rate?

WORK IT OUT >>

Question 6 - Answer

- What other sources of data might you use to evaluate whether or not the infant mortality initiative reduced the infant mortality rate?



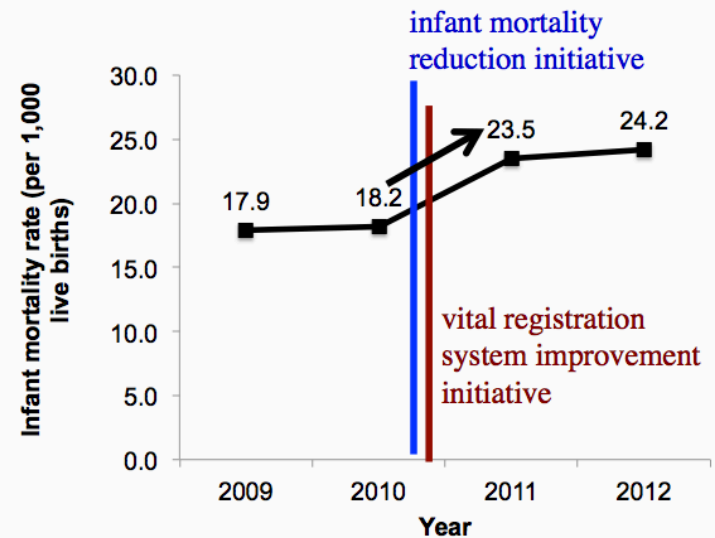
Question 7 - Question

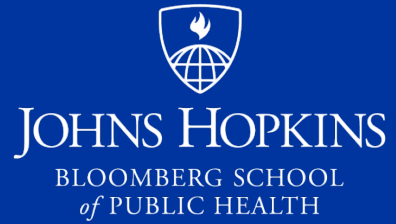
- How will improvements and changes in the vital registration system impact your assessment of the infant mortality rate over time?

WORK IT OUT >>

Question 7 - Answer

- How will improvements and changes in the vital registration system impact your assessment of the infant mortality rate over time?





SECTION E: Public Health Information Systems: Population-based Surveys

Public Health Information Systems (PHIS)

- Traditional sources
 - ▶ Surveys that are population-based and topic-oriented
 - ▶ Registries of specific health outcomes
 - ▶ Surveillance systems
- Provide data for health statistics beyond mortality and routine statistics
- These data are richer for their intended purposes compared to civil/vital registry data because they are collected for a (usually specific) public health purpose

Population-based Surveys

- Local: In Baltimore
 - ▶ Baltimore City Community Health Survey (CHS)
- National: In the United States
 - ▶ National Health and Nutrition Examination Survey (NHANES)
- International: Surveys used by many countries
 - ▶ World Health Survey
 - ▶ Demographic and Health Surveys (DHS)
- These surveys measure self-reported health behaviors
 - ▶ Some also take specimens

Baltimore City Community Health Survey, 2014: <http://health.baltimorecity.gov/sites/default/files/BCHD%20CHS%20Report%20Sept%2016%202015.pdf>

NHANES: <http://www.cdc.gov/nchs/nhanes/index.htm>

WHO World Health Survey: <http://www.who.int/healthinfo/survey/en/>

Demographic and Health Surveys: <http://dhsprogram.com>

Baltimore City Community Health Survey (2014) 1

- Description: Objectives of the survey:
 - ▶ Assess the health status and needs of residents
 - ▶ Identify gaps in access to health services
 - ▶ Assess the use and perception of the Affordable Care Act and the Maryland Health Exchange
 - ▶ Assess the use and perception of city health services



Baltimore City Community Health Survey, 2014: <http://health.baltimorecity.gov/sites/default/files/BCHD%20CHS%20Report%20Sept%202016%202015.pdf>

Baltimore City Community Health Survey (2014) 2

- Design:
 - ▶ Cross-sectional telephone interview
 - ▶ 1,722 participants
- Sampling design:
 - ▶ Random-digit dialing, computer-assisted telephone interview survey
 - Landlines and cell phone
 - ▶ Adults only, no over-sampling of any specific population
 - ▶ N=1,722 respondents
 - ▶ Sampling weights
 - Weights reflect the sex, age, and race/ethnic characteristics of the city



NHANES 1

- Description: To assess the health and nutritional status of adults and children in the US
- Design:
 - ▶ Cross-sectional survey with a clinical exam, lab tests, interview
 - ▶ 5,000 people from 15 selected counties



NHANES 2

- Sampling design:
 - ▶ Counties are the primary sampling units
 - ▶ Within each county, clusters of households are selected
 - ▶ One or more persons per household are selected for the sample
 - ▶ Sampling weights are used to correct unequal probabilities of selection and non-response adjustments
 - ▶ It is recommended that two, two-year cycles are included in a single study for stability of the estimates



World Health Survey 1

- Description: To strengthen capacity to monitor critical health outcomes and health systems
 - ▶ 70 participating countries
 - ▶ 300,000 total sample size
- Design:
 - ▶ Household face-to-face interviews or computer-assisted telephone interview



World Health Survey 2

- Sampling:
 - ▶ Sampling frame must cover 100% of the eligible population - cannot restrict by ethnicity, geography, etc.
 - ▶ Single stage random sample is ideal
 - Multi-stage cluster sampling is an option
 - ▶ Sampling weights are used to correct unequal probabilities of selection and non-response adjustments

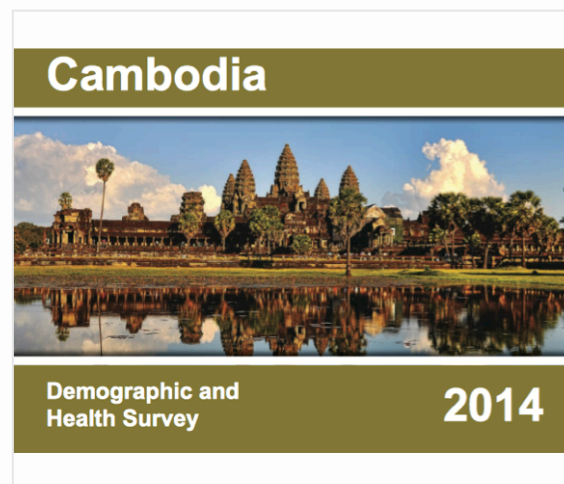


Demographic and Health Surveys 1

- DHS take on average 18-20 months to complete
- Four phases:
 - ▶ Survey preparation, (including sampling design) and questionnaire design
 - ▶ Training and fieldwork
 - ▶ Data processing
 - ▶ Final report, data preparation and dissemination

Demographic and Health Surveys 2

- Data sets, tools, and reports are available online
- Cambodia, 2014
 - ▶ Total fertility rate (children per women) = 2.7
 - ▶ 56% of currently married women use a method of contraception
 - ▶ Infant mortality rate = 28/1,000 live births
 - ▶ 73% of children under age 5 have registered births
 - ▶ 73% of children age 12-23 months are fully immunized



More Demographic and Health Surveys

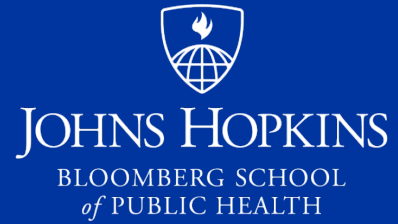
- DHS administers 260 surveys in >90 countries
 - ▶ Demographic and Health Surveys
 - Obtain data for monitoring and impact evaluation indicators
 - ▶ AIDS Indicator Surveys
 - Obtain indicators for effective monitoring of national HIV/AIDS programs
 - ▶ Malaria Indicator Surveys
 - Obtains indicators on malaria prevention and treatment
 - ▶ Service Provision Assessment Surveys
 - Characteristics of health facilities and services available
 - ▶ Key Indicator Surveys
 - Monitoring and evaluation data for population and health activities in small regions that may be targeted by an individual's project
 - ▶ Other quantitative surveys
 - ▶ Qualitative research

Limitations to Population-based Survey 1

- Design
 - ▶ No longitudinal follow-up
- Measurement
 - ▶ Questions have to be acceptable to the culture
 - ▶ Recall bias/social desirability bias
 - ▶ Missing data
 - ▶ Near impossible to confirm reported health events

Limitations to Population-based Survey 2

- Study population
 - ▶ Sampling methodology can get complicated very quickly
 - ▶ Biases associated with leaving groups out (cell phones, institutionalized, ill, race/ethnicity, etc.)
- Analyses
 - ▶ Weights must be appropriately calculated and applied



SECTION F: Public Health Information Systems: Registries

Registries

- Registries are data collected on a specific disease outcome
 - ▶ Registries only include “cases”
 - ▶ Registries are designed for specific outcomes, so the data can be rich
- Examples
 - ▶ International Agency for Research on Cancer (IARC) www.iarc.fr
 - ▶ European Surveillance of Congenital Anomalies (Eurocat)
<http://www.eurocat-network.eu/>
 - ▶ United States Renal Data System (USRDS)
<http://www.usrds.org/>

International Agency for Research on Cancer

- Description:
 - ▶ IARC cultivates international collaboration in cancer research
- Design:
 - ▶ Registries for more than 200 type-specific cancers
- Resources:
 - ▶ Global Cancer Observatory: an interactive web-based platform presenting global cancer statistics
 - ▶ World Cancer Report
 - ▶ WHO/IARC Classification of Tumors (“The blue book”)
 - ▶ Biobank of specimens

International Agency for Research on Cancer



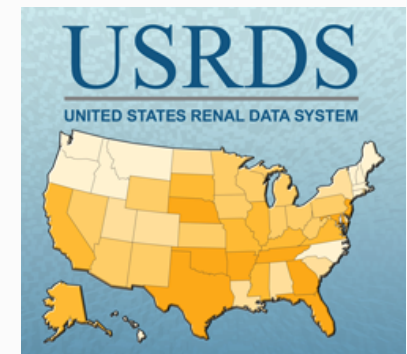
European Surveillance of Congenital Anomalies

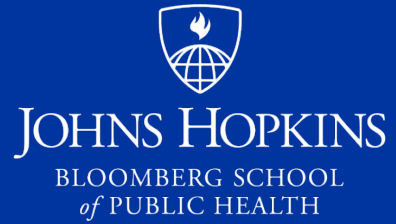
- Description: A network of population-based registries of congenital anomalies
- Design
 - ▶ 43 registries in 23 countries
 - ▶ Standard data entry and coding systems used across the registries
- Resources
 - ▶ Bi-annual update of prevalence data tables
 - ▶ Data made available to researchers after approval
 - ▶ Key public health statistics, such as perinatal mortality



United States Renal Database System

- Description: National resource for data about chronic kidney disease and end-stage renal disease in the US
- Design:
 - ▶ Renal replacement therapy, specifically dialysis, is paid for in the US through Medicare & Medicaid Services (CMS)
 - ▶ This database is created through a partnership of the National Institutes of Health and CMS near complete capture of all people with end-stage renal disease in the US
- Resources:
 - ▶ Annual Data Report
 - ▶ Standard Analysis Files
 - ▶ Data can be shared (\$)





SECTION G: Public Health Information Systems: Surveillance Systems

Surveillance

- Surveillance is
 - ▶ “Systematic and continuous collection, analysis, and interpretation of data, closely integrated with the timely and coherent dissemination of the results an assessment to those who have the right to know so that action can be taken. It is an essential feature of epidemiological and public health practice.” (Porta 2014)
- Example of surveillance data
 - ▶ FluView: CDC’s Influenza Surveillance

Surveillance

- Description: Nearly real-time (2 week delay) virologic and clinical surveillance of influenza in the US
- Design: a collaboration of
 - ▶ Virologic surveillance of public health and clinical laboratories in all 50 states, Puerto Rico, and District of Columbia
 - ▶ Pneumonia and influenza-associated deaths through the Mortality Surveillance System and the 122 Cities Mortality Reporting System (NCHS)
 - ▶ Influenza-related hospitalizations: Influenza Hospitalizations Surveillance Network (FluSurv-NET)
 - ▶ Influenza-like illness reported through the US Outpatient Influenza-like Illness Surveillance Network (ILINet)
- Resources:
 - ▶ <http://www.cdc.gov/flu/weekly/index.htm>



SECTION H: Common Problems of Routine and Public Health Information Systems

Common Problems of RHIS and PHIS I

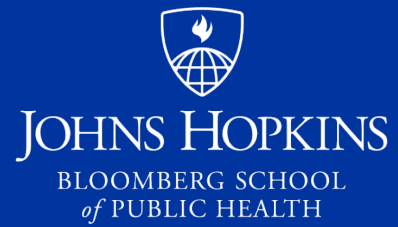
- The majority of these systems were developed independent of one another
 - ▶ Combining data across systems can be logistically prohibitive
- Lack of adequate technical guidelines
- Lack of trained human resources

Common problems of RHIS and PHIS II

- Infrastructure is inadequate in many countries
- Single-disease oriented systems
- Information on interventions is not linked
- “Real-time” data is rare - most have a multi-year lag

RHIS and PHIS - What Do You Do With the Data?

- Health statistics
 - ▶ Aggregated data describing and enumerating attributes, events, behaviors, services, resources, outcomes, or costs related to health, disease, and health services. Data may be derived from survey instruments, medical records, and administrative documents
- Health indicators
 - ▶ A variable, susceptible to direct measurement, that reflects the state of health of persons in a community



Summary

Summary

- Routine information systems include
 - ▶ **Civil** registration and **vital statistics** systems
 - ▶ **Medical** record systems
 - ▶ **Administrative** record systems
- Public health information systems include
 - ▶ **Surveys** that are population-based and topic-oriented
 - ▶ **Registries** of specific health outcomes
 - ▶ **Surveillance**
- Routine and public health information systems are of varying quality
 - ▶ These systems provide important information
 - ▶ These systems have challenges
- Assessing the quality of mortality data is one example of the importance of understanding the quality of the data